

City Scale Retrofitting Analysis – Ahmedabad

Organization: SWA Consultancy

Intern: Kavya Durga Sri Gullapalli

Week: Week 1

Presented by:

Kavya Durga Sri



AFTER RETRO-FITTING ATTIC



BEFORE RETRO-FITTING ATTIC

Introduction & Project Understanding

What I Learned This Week:

- Attended the introductory internship meeting and understood the overall project workflow.
- Learned the basics of green buildings, sustainability, and retrofit analysis.
- Understood how buildings gain heat and how insulation helps reduce cooling load.
- Studied thermal concepts such as heat transfer, U-value, R-value, thermal conductivity, and vapour resistance.
- Learned the importance of energy-efficient building envelopes in hot climates like Ahmedabad.





Insulation & Thermal Performance

Thermal Insulation Concepts

Topics Studied

- Modes of heat transfer:
 - Conduction
 - Convection
 - Radiation
- Insulation properties:
 - Thermal conductivity
 - Thermal resistance (R-value)
 - U-value
 - Thermal diffusivity
 - Vapour resistance

Insulation Materials

- Natural materials:
 - Glass wool
 - Mineral wool
 - Cellulose
- Synthetic materials:
 - EPS
 - XPS
 - PUF
 - PIR

Key Understanding

- Insulation works mainly because of trapped air/gas pockets.
- Closed-cell insulation provides better thermal and moisture resistance.

Green Building Concepts & Building Codes

IGBC & GRIHA

- Understood green building rating systems used in India.
- Learned about energy efficiency, water conservation, sustainable materials, and indoor environmental quality.

ECBC 2017

- Learned about Energy Conservation Building Code standards.
- Studied concepts related to:
 - Building envelope
 - U-value
 - SHGC
 - HVAC efficiency
 - Energy-efficient design

Passive Design Strategies

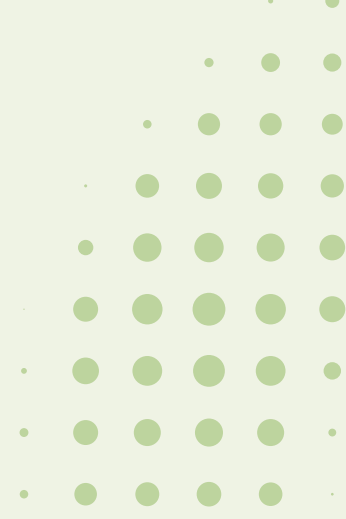
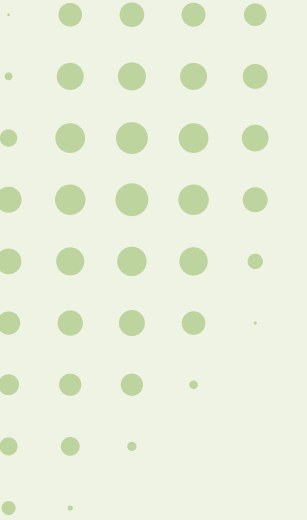
- Roof insulation
- Shading
- Reflective surfaces
- Daylighting
- Natural ventilation

CityBES Exploration & Next Steps

Initial Exploration of CityBES

- Explored the CityBES platform and gained a preliminary understanding of its purpose and workflow in urban building energy analysis.
- Learned that CityBES is an Urban Building Energy Modeling (UBEM) platform used for analyzing building energy performance and retrofit scenarios at city scale.
- Developed an initial understanding of how retrofit analysis can help evaluate energy savings before and after building interventions.

- Learn the dataset preparation workflow for Ahmedabad buildings
- Study how datasets can be prepared using QGIS
- Explore retrofit scenario implementation in CityBES
- Understand building energy simulation workflows

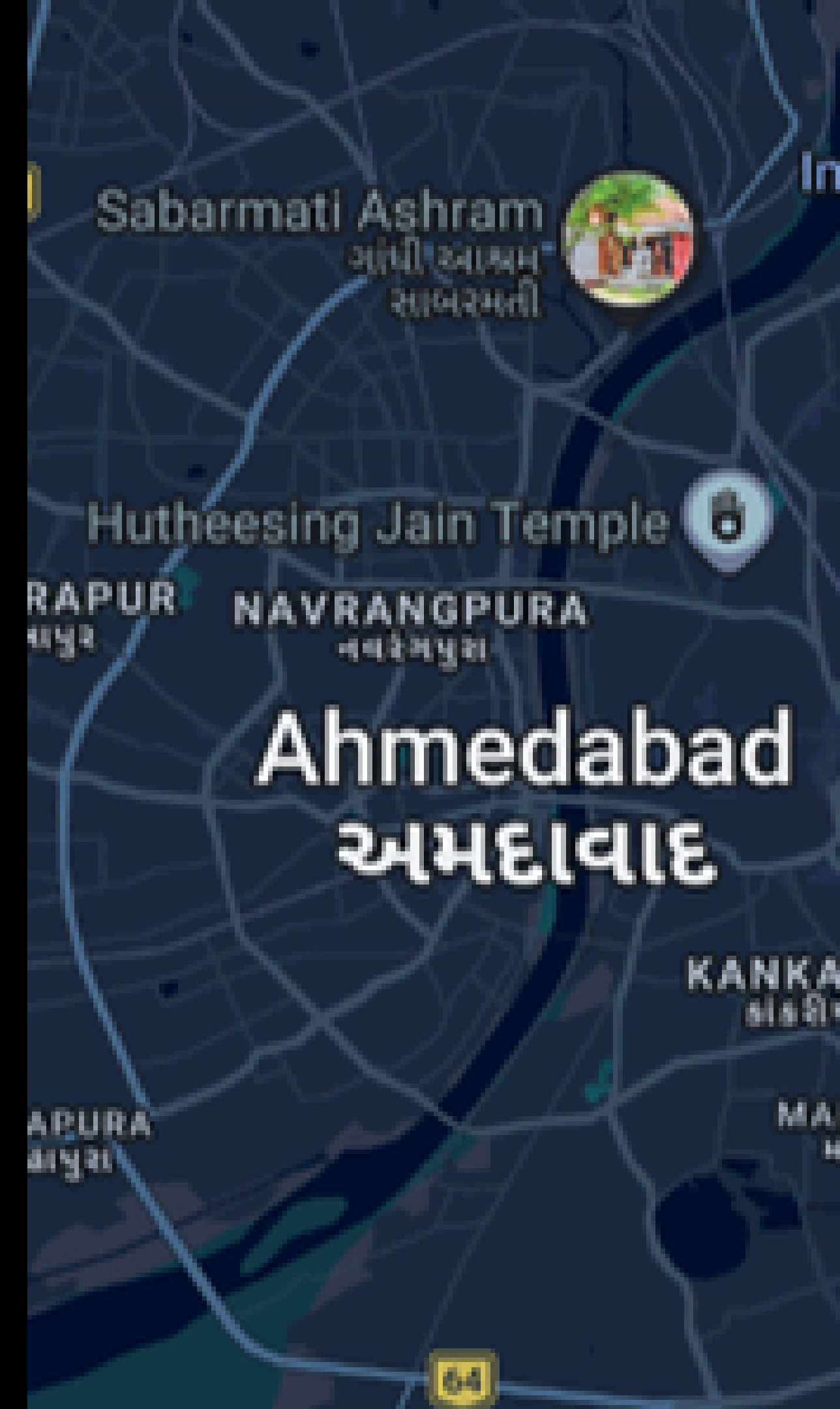


Thank you

CITY SCALE RETROFITTING ANALYSIS – AHMEDABAD

Organization: SWA Consultancy
Intern: Kavya Durga Sri Gullapalli
Week: Week 2

➤➤➤➤ Presented by Kavya Durga Sri



INTERNSHIP LEARNING FOCUS THIS WEEK



During this week, I focused on understanding the practical workflow involved in Urban Building Energy Modeling (UBEM) and dataset preparation for CityBES.

The learning mainly involved:

- Exploring CityBES platform and its workflow
- Understanding GIS-based urban energy modeling
- Studying satellite imagery sources
- Learning the basics of QGIS and spatial datasets
- Understanding how building datasets are prepared for energy simulations

UNDERSTANDING CITYBES PLATFORM



Explored the CityBES platform and gained understanding of:

- District-level building visualization
- Urban Building Energy Modeling (UBEM)
- Building energy simulations
- Building archetypes and datasets
- Baseline energy simulations
- Energy performance comparison methods

Important Features Explored

- Building filters
 - Color buildings by energy metrics
 - Result options and baseline simulations
 - District buildings and urban environment modules
- 

Key Understanding

CityBES is used to convert city-scale building datasets into digital energy models for analyzing building energy performance and retrofit scenarios.



Show Building Filters

Hide Building Coloring Options

Parameter

Options

Result options

Baseline Simulated

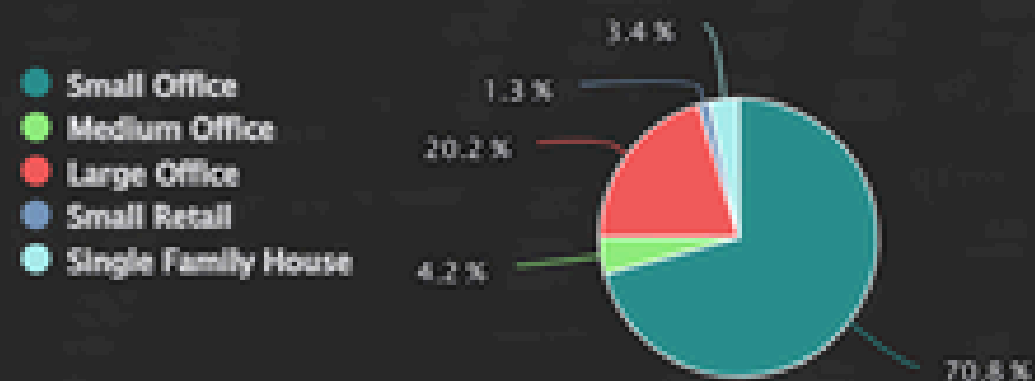
Color buildings by

Site Energy Use Intensity

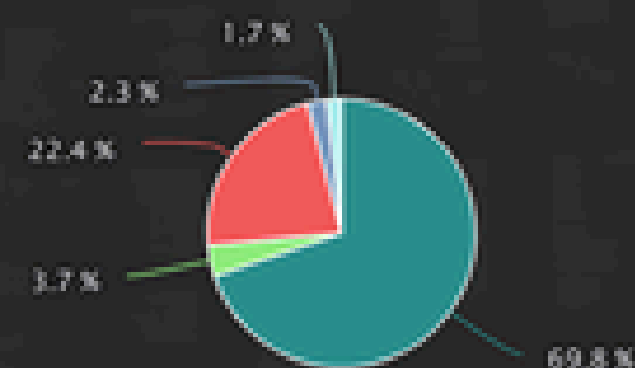
Show summary by

Current Result Option

Building Floor Area

Total: 65,697 (m²)

Site Energy Use



Total: 13 (GWh)

Site Energy Use Intensity
Unit: kWh/m²

5655 GEARY BLVD

Edit

Name	5655 GEARY BLVD
Building Type	Small Office
Year Built	1977
Number of Stories	2
Building Height	9 m
Total Floor Area	1074 m ²

Baseline Simulated Results

Site Energy Use Intensity	192 kWh/m ²
Source Energy Use Intensity	536 kWh/m ²
Operational GHG Emission Intensity	34 kg.CO ₂ /m ²
Peak Electricity Load Intensity	56 W/m ²
Electricity Use Intensity	160 kWh/m ²
Natural Gas Use Intensity	31 kWh/m ²
Cooling Electricity Use Intensity	0 kWh/m ²
Heating Site Energy Use Intensity	0 kWh/m ²
Internal Lighting Electricity Use Intensity	0 kWh/m ²
Internal Equipment Electricity Use Intensity	0 kWh/m ²
Service Hot Water Site Energy Use Intensity	0 kWh/m ²



SATELLITE DATA EXPLORATION

Explored multiple satellite imagery and GIS data sources for Ahmedabad urban analysis:

- Google Earth Pro
- VEDAS
- Bhuvan NRSC
- Copernicus Browser

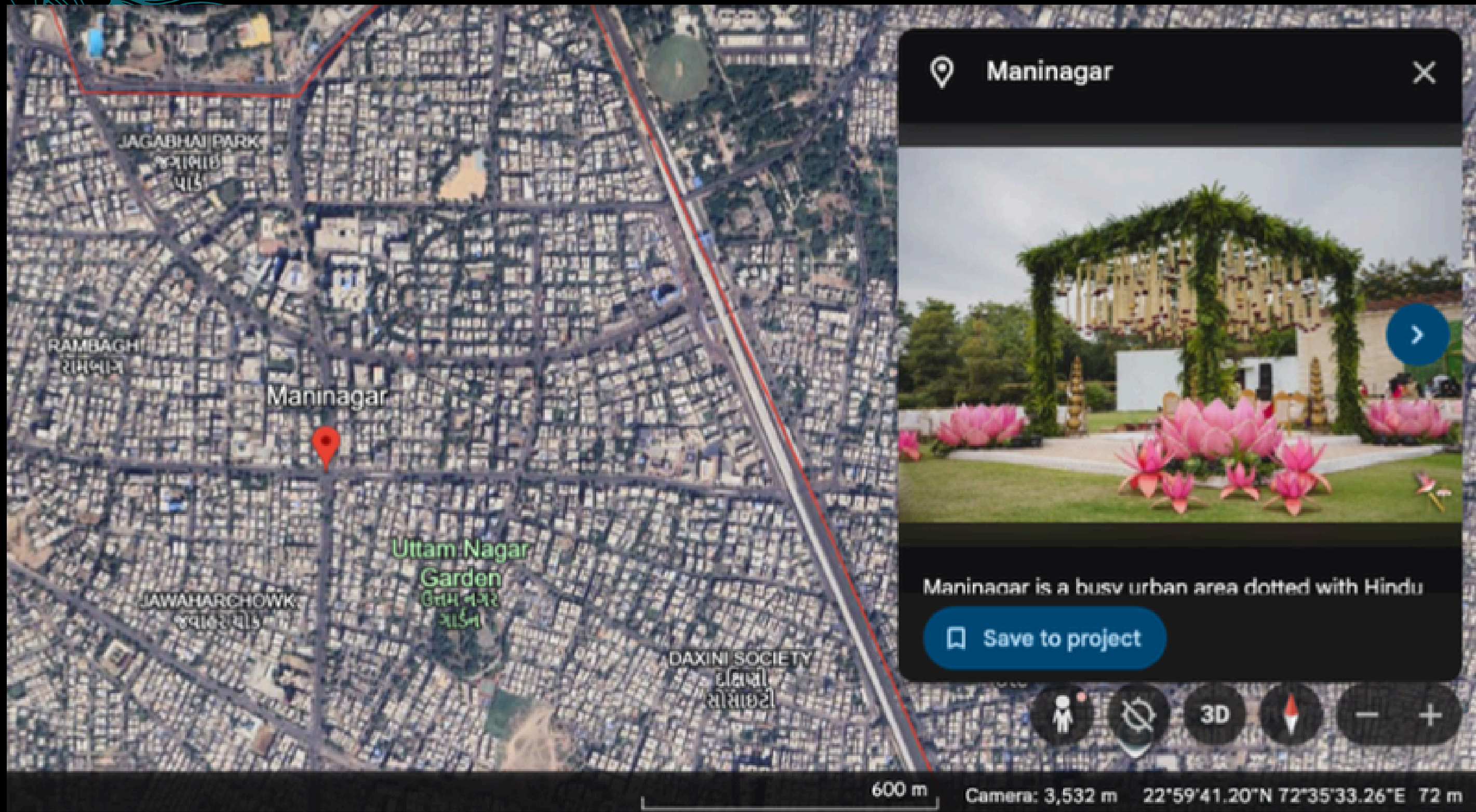
Collected satellite imagery and screenshots

Learned to identify exact locations using latitude, longitude, and elevation data

Key Understanding

Satellite imagery is important for understanding urban characteristics before preparing GIS datasets and energy models.





Selected Ahmedabad Areas

- Navrangpura
- Maninagar
- Makabra
- Bopal
- Naroda
- Chandkheda
- Vastrapur



DATASET PREPARATION WORKFLOW & QGIS LEARNING

Understanding Dataset Preparation Workflow

Learned the workflow required for preparing datasets for CityBES simulations.

Workflow Studied

1. Collect satellite and GIS data
2. Import spatial data into QGIS
3. Create building footprint polygons
4. Add building attributes:
 - Building height
 - Number of stories
 - Building type
 - Year built
5. Export datasets as GeoJSON
6. Upload datasets into CityBES

QGIS Exploration:

Started exploring QGIS and understood:

- GIS layers
- Building footprints
- Spatial datasets
- GeoJSON format
- Coordinate systems and CRS

Key Understanding

QGIS is used for preparing spatial building datasets, while CityBES is used for building energy simulations.





NEXT STEPS

- Continue learning QGIS workflows
- Understand building footprint extraction
- Learn GeoJSON dataset creation in detail
- Explore Ahmedabad building dataset preparation
- Study retrofit scenario implementation in CityBES

Overall Understanding:

The overall workflow involves converting real-world urban buildings into digital building energy models for retrofit and sustainability analysis.





**THANK
YOU**

City Scale Retrofitting Analysis for Ahmedabad

Week 3 Progress Presentation

Intern: Kavya Durga Sri

Organisation: SWA Consultancy

Objectives for This Week

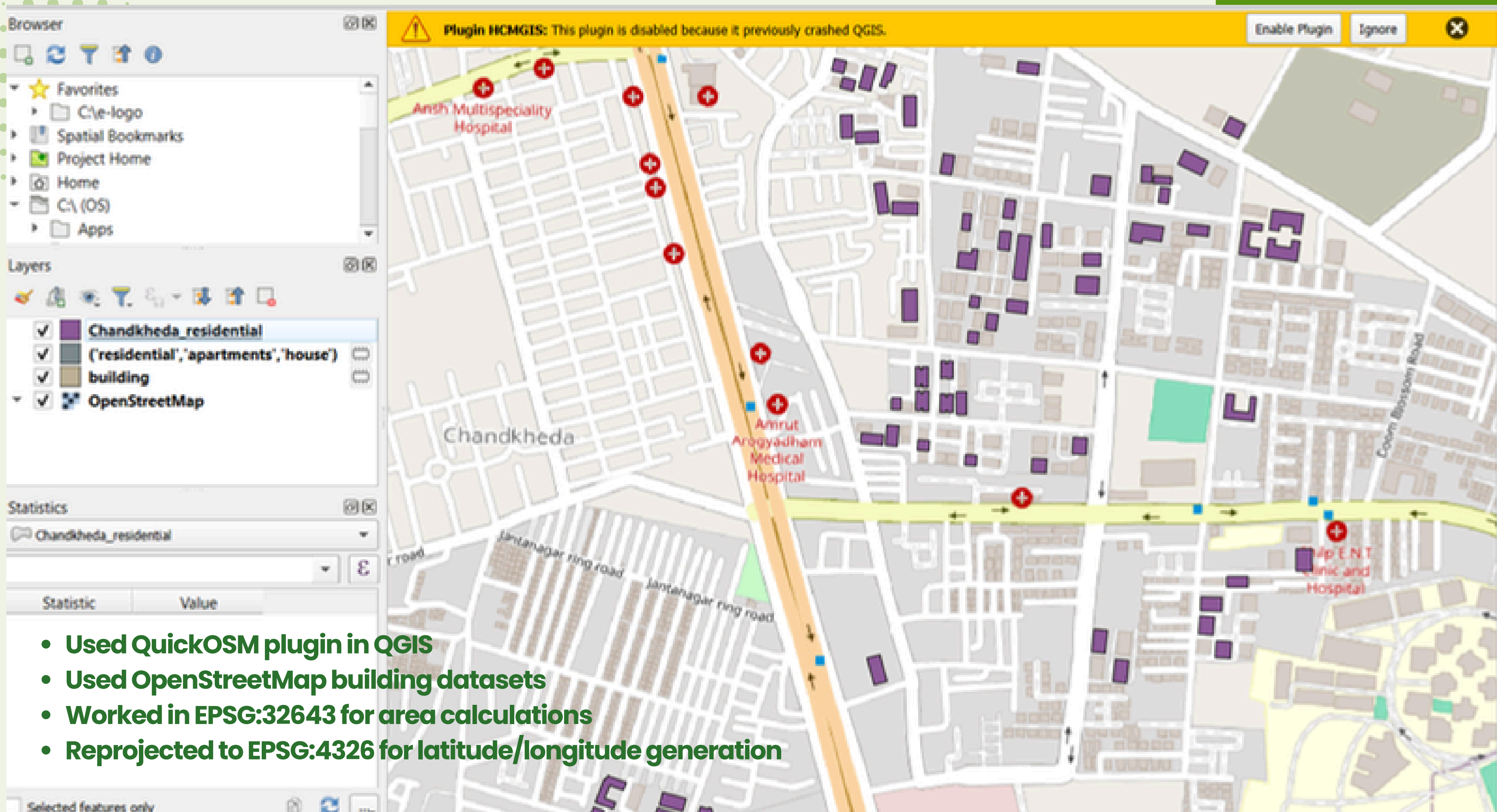
- Understand GIS-based dataset preparation workflow for UBEM
- Explore building footprint extraction using QGIS
- Prepare residential building datasets for selected Ahmedabad regions
- Estimate building heights and building attributes
- Understand spatial divisions and GeoJSON preparation for CityBES
- Explore urban GIS portals and remote sensing platforms

Areas Selected

- Vastrapur
- Navrangpura
- Naroda
- Chandkheda
- Bopal

Building Dataset Preparation Workflow





[illegible]

Methods Explored

- OpenStreetMap building:levels
- Google Earth Pro visual verification
- VEDAS and AUDA Geoportal exploration

Final Practical Workflow :



Key Understanding

- Approximate heights are acceptable for preliminary UBEM datasets
- Exact engineering-grade heights are not mandatory at city scale
- Different building typologies were identified visually

Challenges

- Missing floor information in some areas
- Blurry 3D visualization in Google Earth Pro
- Difficulty in obtaining exact heights from public datasets

Browser

✖

📁 🔄 🔍 📌 📍

★ Favorites

- 📁 C:\e-logo
- 📁 Spatial Bookmarks
- 📁 Project Home
- 📁 Home
- 📁 C:\OS
- 📁 Apps

Layers

✖

📁 🔄 🔍 📌 📍

- ✓ ■ Bopal
- ✓ ■ ('residential', 'apartments', 'house')
- ✓ ● building
- ✓ ■ OpenStreetMap

Statistics

✖

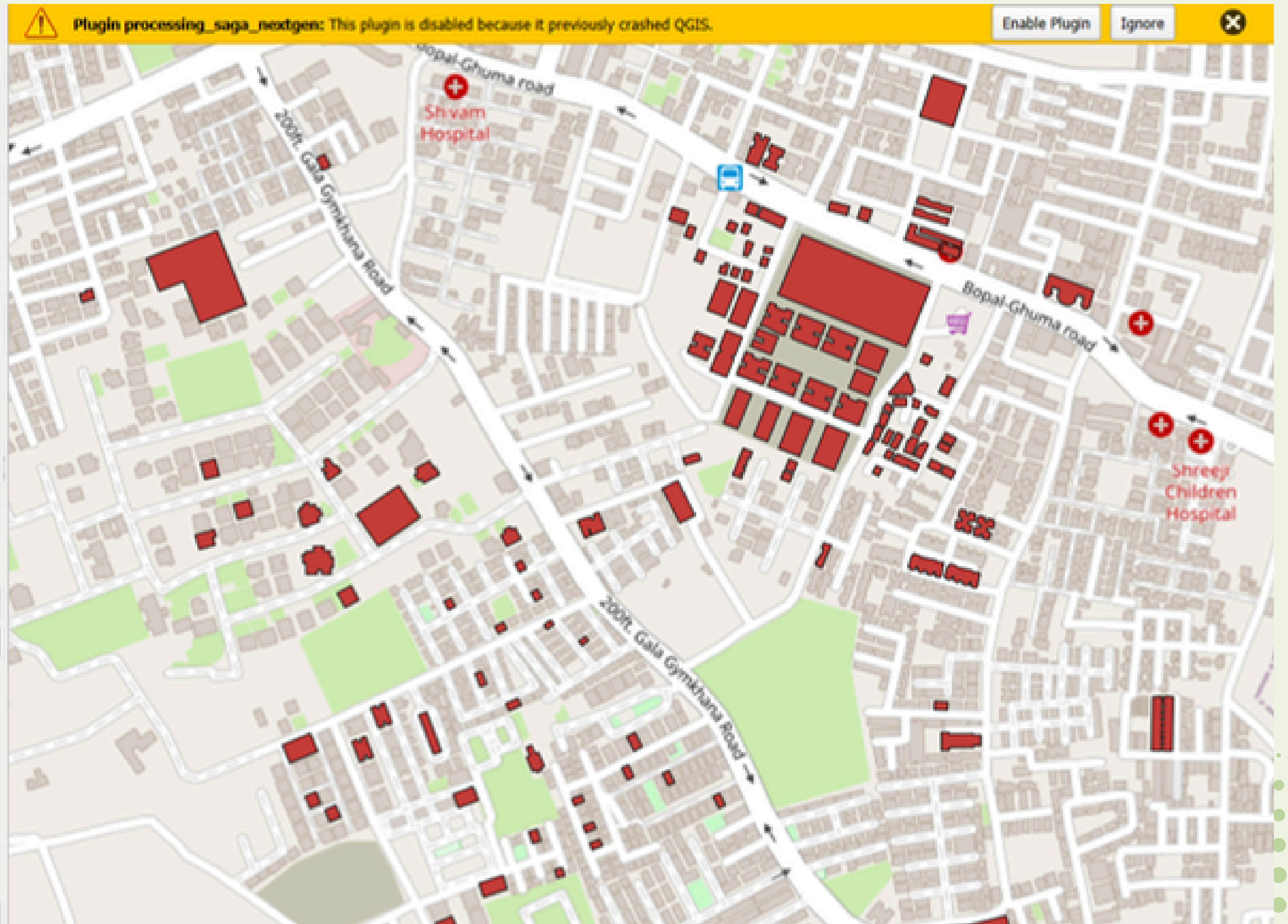
building

📊

Statistic	Value
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☐ Selected features only

📁 🔄 🔍 📌 📍

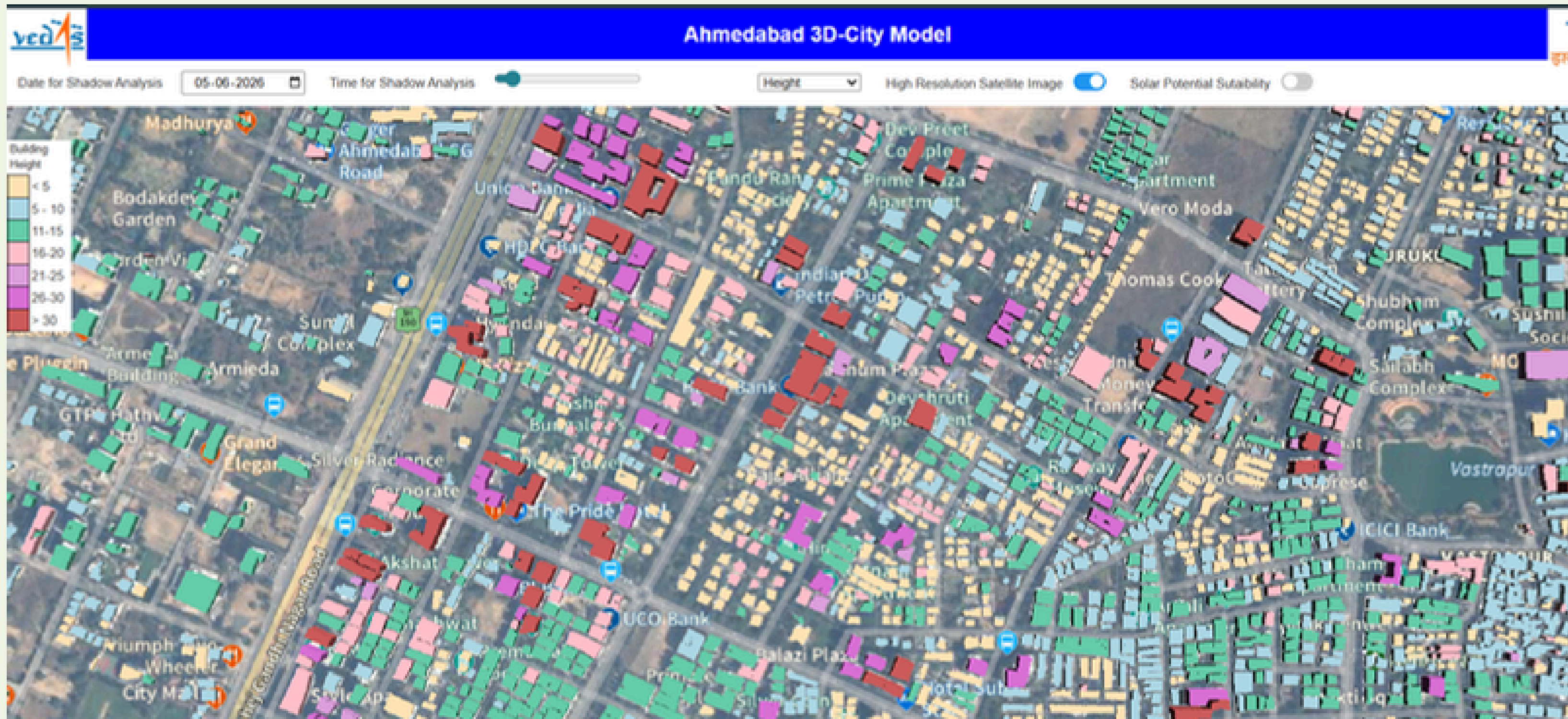


[illegible]

roof:shape	roof:material	building:material	building:levels	building:colour	Area	Height	latitude	longitude
flat	concrete	plaster;glass	10	beige	956.931	30	23.041	66.531
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flat	concrete	plaster;glass	10	beige	956.94	30	23.042	66.531
flat	concrete	plaster;glass	10	beige	956.945	30	23.041	66.532
flat	concrete	plaster;glass	10	beige	957.098	30	23.042	66.532
flat	concrete	plaster;glass	4	beige	281.562	12	23.043	66.531
flat	concrete	plaster	5	beige	573.019	15	23.037	66.525
flat	concrete	plaster;glass	2	beige	315.914	6	23.042	66.53
flat	concrete	plaster	8	beige	374.711	24	23.044	66.529
flat	concrete	plaster	5	beige	395.127	15	23.04	66.527
flat	concrete	plaster	10	pink	413.307	30	23.04	66.527
flat	concrete	plaster	10	beige	370.81	30	23.039	66.527
flat	concrete	plaster	10	beige	642.352	30	23.039	66.527
flat	concrete	plaster	10	beige	373.238	30	23.039	66.527
flat	concrete	plaster	10	beige	515.849	30	23.039	66.527
flat	concrete	plaster;glass	15	beige	559.647	45	23.041	66.53
flat	concrete	plaster	5	beige	387.661	15	23.04	66.522
flat	concrete	plaster	14	white	918.852	42	23.042	66.535

Obtained Latitude, Longitude information and added to the attribute tables through expression

Explored the Ahmedabad 3D City Model platform available through VEDAS/ISRO to understand building height distribution and urban morphology in Ahmedabad. The platform provides color-coded visualization of buildings based on height ranges along with shadow analysis and solar potential analysis. This was explored as an additional method for approximate building height estimation and validation for selected study areas.



Next Steps

- Complete dataset preparation for all selected Ahmedabad regions.
- Improve building height estimation for areas with missing floor information.
- Explore methods for obtaining more accurate building attributes using GIS and remote sensing platforms like AMC Citymap, vedas 3d city model
- Divide datasets based on urban regions/planning subdivisions as recommended in the CityBES workflow.
- Understand attribute joining and spatial clipping workflows in QGIS.
- Should add year built and other data which is not available in attribute tables
- Learn the complete CityBES dataset upload and mapping process.



**THANK
YOU**



City Scale Retrofitting Analysis for Ahmedabad

Organization: SWA Consultancy

**Interns: Kavya Durga Sri Gullapalli
Srushti Dandekar**

PROJECT OVERVIEW & OBJECTIVES

This internship focuses on understanding sustainable urban development through GIS-based building analysis and energy retrofitting workflows.

The work that will be done

- Understanding green building concepts
- Learning Urban Building Energy Modeling (UBEM)
- Preparing GIS datasets using QGIS
- Exploring CityBES energy simulation workflows
- Studying retrofit strategies for energy-efficient buildings

Objectives

- Understand GIS and remote sensing workflows
- Learn building energy concepts and retrofit analysis
- Prepare Ahmedabad building datasets
- Explore QGIS and CityBES workflows
- Study sustainable building standards and urban energy analysis

FUNDAMENTAL CONCEPTS

Building Insulation

- Open-cell and closed-cell insulation
- Moisture resistance
- Thermal insulation materials
- Impact of insulation on cooling load reduction

Green Building & Retrofitting

- Green building principles
- Passive cooling strategies
- Building envelope optimization
- Retrofit measures for improving energy efficiency

Thermal Performance Parameters

- U-value and R-value
- Thermal conductivity
- Heat flux
- Vapour resistance



AFTER RETRO-FITTING ATTIC

Modes of Heat Transfer

- Conduction through walls, roofs and floors
- Convection through air movement
- Radiation from solar heat gain



BEFORE RETRO-FITTING ATTIC

Standards Explored

ECBC 2017 (Energy Conservation Building Code)

- Provides guidelines for improving building energy efficiency through optimized building envelopes, HVAC systems, lighting, and electrical systems.

IGBC (Indian Green Building Council)

- A green building rating system that evaluates sustainability aspects such as energy efficiency, water conservation, indoor environmental quality, and resource management.

These standards provide practical guidelines and evaluation frameworks for designing energy-efficient and sustainable buildings, which form the foundation for retrofit strategies and green building development.

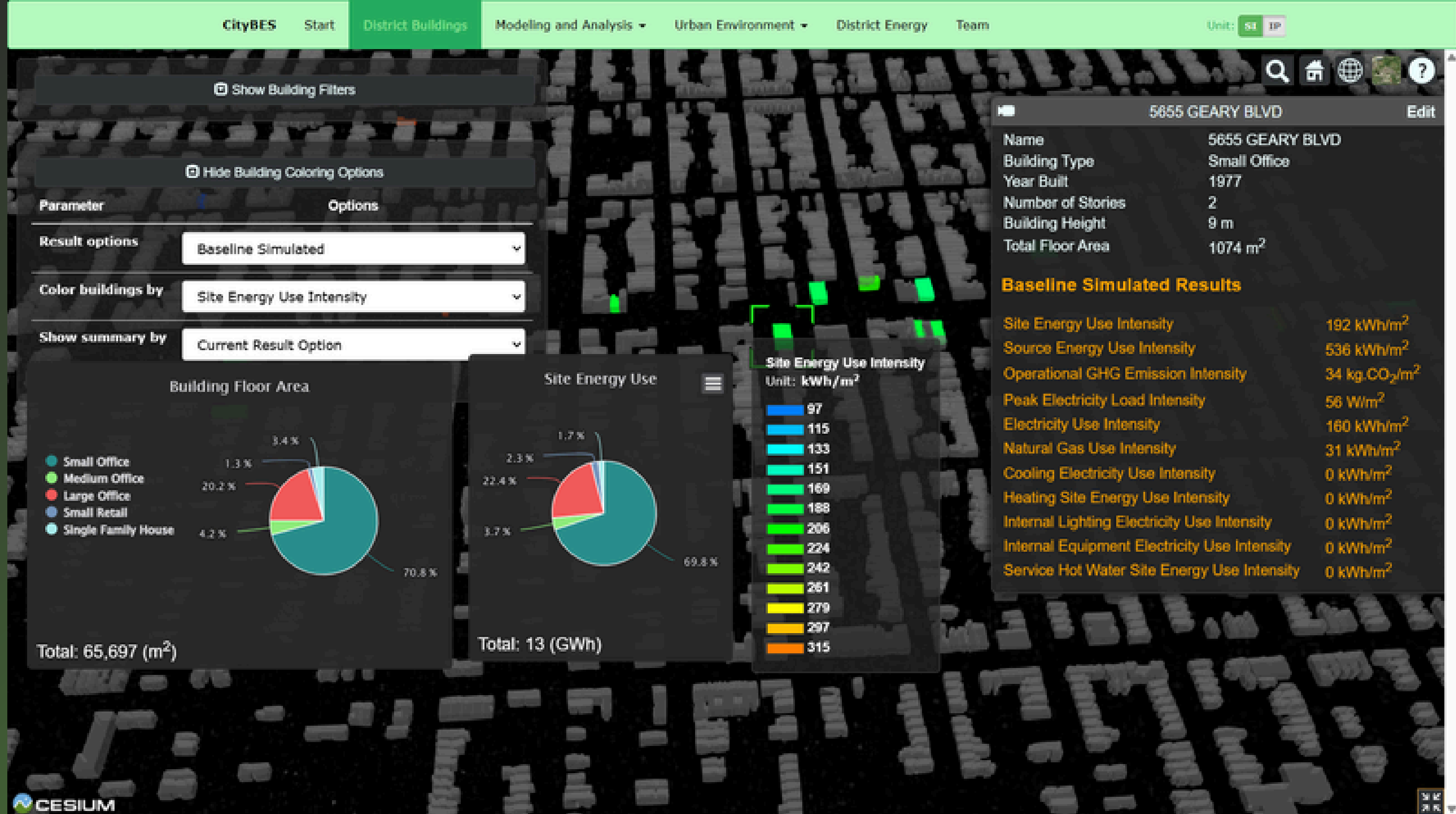
CITYBES & SATELLITE DATA EXPLORATION

Features Explored

- District Building Visualization
- Building Information & Attributes
- Building Filtering and Classification
- Baseline Energy Simulation Results
- Building Energy Performance Comparison
- Retrofit Analysis Workflows
- Urban Building Energy Modeling (UBEM)
- City-Scale Building Energy Assessment

Understanding Gained

- CityBES is a platform for city-scale building energy analysis.
- Building datasets can be converted into digital energy models.
- Baseline simulations help evaluate building energy performance.
- Retrofit scenarios can be used to estimate potential energy savings.
- Building attributes influence overall energy performance.



CityBES Interface Showing Baseline Building Energy Simulation Results and City-Scale Energy Performance Visualization

GIS DATASET PREPARATION WORKFLOW

Workflow Followed



Tools Used

- QGIS
- QuickOSM Plugin
- OpenStreetMap
- Google Earth Pro
- Ahmedabad 3D City Model

Platforms Explored

- CityBES
- Google Earth Pro
- VEDAS
- Bhuvan NRSC
- Copernicus Browser

QGIS-Based Building Extraction & Spatial Processing

Residential building datasets were prepared using OpenStreetMap data and QGIS spatial processing workflows.

Activities Performed

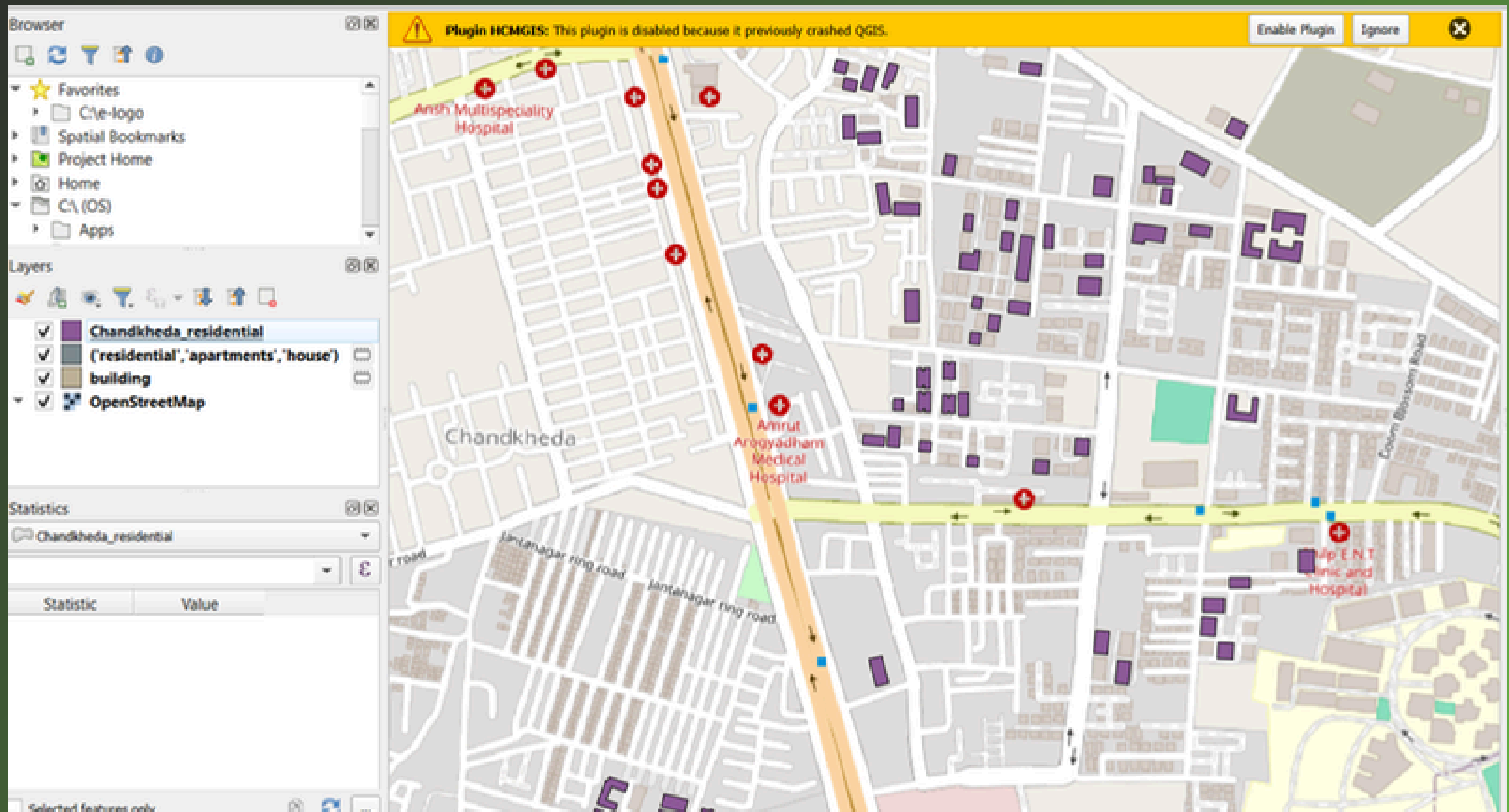
- Extracted building footprints using QuickOSM
- Filtered residential buildings
- Selected representative building samples
- Calculated building areas
- Generated latitude and longitude attributes
- Worked with CRS transformations

Coordinate Systems Used

- EPSG:32643 → Area calculations
- EPSG:4326 → Latitude/Longitude generated

Output Generated

- Residential building datasets
- Building footprint maps
- Spatial attribute tables

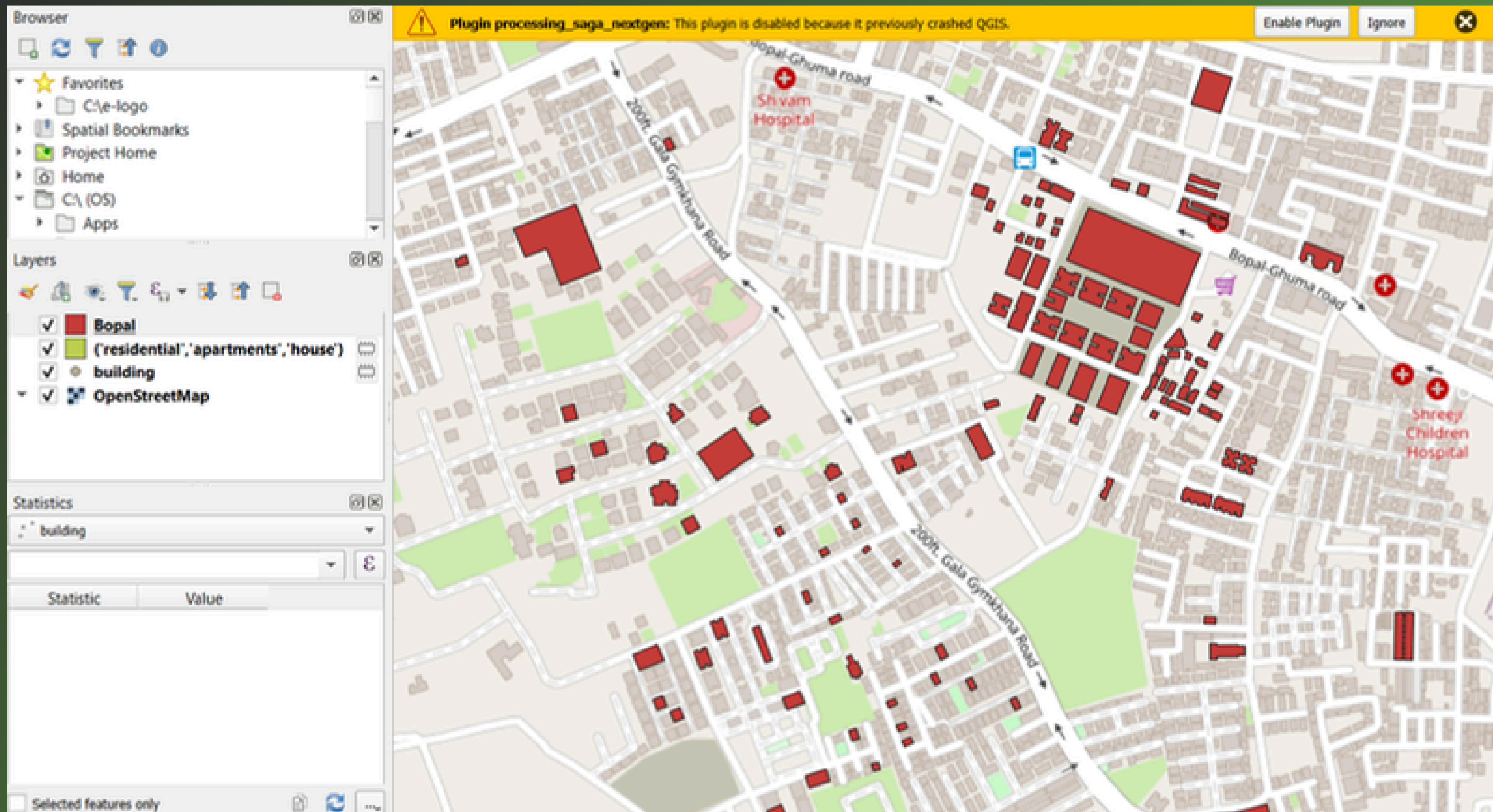


Residential Building Mapping and Visualization in QGIS for Chandkheda, Ahmedabad

Chandkheda_residential — Features Total: 61, Filtered: 61, Selected: 0

	religion	amenity	tourism	name	addrpostcode	addrcity	roofshape	roofmaterial	buildingmaterial	buildinglevels	buildingcolour	natural	type	Area_building	Height
1	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	525.668	18
2	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	pink	NULL	NULL	337.409	18
3	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	8	beigebrown	NULL	NULL	491.784	24
4	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	263.27	18
5	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	8	beigebrown	NULL	NULL	685.198	24
6	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beigebrown	NULL	NULL	459.49	18
7	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	14	beigebrown	NULL	NULL	796.562	42
8	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	9	beige	NULL	NULL	665.832	27
9	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	9	beige	NULL	NULL	665.866	27
10	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	556.668	18
11	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	white	NULL	NULL	769.19	18
12	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	1277.218	18
13	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	315.614	18
14	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	502.253	18
15	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	489.037	18
16	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	444.347	18
17	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	723.199	18
18	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	453.18	18
19	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	473.134	18
20	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	636.74	18
21	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	237.693	18
22	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	394.696	18
23	NULL	NULL	NULL	NULL	NULL	NULL	flat	concrete	plaster	6	beige	NULL	NULL	515.34	18

Attribute Database Generated in QGIS



Building Footprint Mapping in QGIS for Bopal, Ahmedabad

Bopal — Features Total: 131, Filtered: 131, Selected: 0

	religion	amenity	layer	sport	smoking	addrstreet	addrpostcode	addrhousenumber	addrcity	natural	buildinglevels	name	leisure	type	Area
1	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	143.253
2	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	89.066
3	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	228.65
4	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	146.532
5	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	271.193
6	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	191.31
7	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	120.061
8	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	592.696
9	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	170.603
10	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	118.046
11	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	park	NULL	7133.047
12	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	334.827
13	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	91.26
14	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	116.809
15	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	117.979
16	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	Gala Aura	NULL	NULL	1169.58
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18	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	116.778
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20	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	638.651
21	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	912.952
22	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	831.916
23	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	1276.016

Show All Features

Attribute Dataset Generated in QGIS for Bopal Region

Obtained Latitude,
Longitude information
and added to the attribute
tables through expression

roof:shape	roof:material	building:material	building:levels	building:colour	Area	Height	latitude	longitude
flat	concrete	plaster;glass	10	beige	956.931	30	23.041	66.531
flat	concrete	plaster	5	lightblue;white	398.256	15	23.037	66.527
flat	concrete	plaster;glass	2	beige	284.885	6	23.044	66.53
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flat	concrete	plaster;glass	10	beige	956.94	30	23.042	66.531
flat	concrete	plaster;glass	10	beige	956.945	30	23.041	66.532
flat	concrete	plaster;glass	10	beige	957.098	30	23.042	66.532
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flat	concrete	plaster	5	beige	387.661	15	23.04	66.522
flat	concrete	plaster	14	white	918.852	42	23.042	66.535

Latitude and Longitude Generation in QGIS Attribute Table

BUILDING HEIGHT ESTIMATION & VALIDATION

Building height is an important attribute required for Urban Building Energy Modeling (UBEM) and CityBES dataset preparation. Since direct height information was not available for all buildings, multiple approaches were explored to estimate and validate building heights.

OSM Building Data



building:levels



Height = Floors × 3 m



Ahmedabad 3D Model

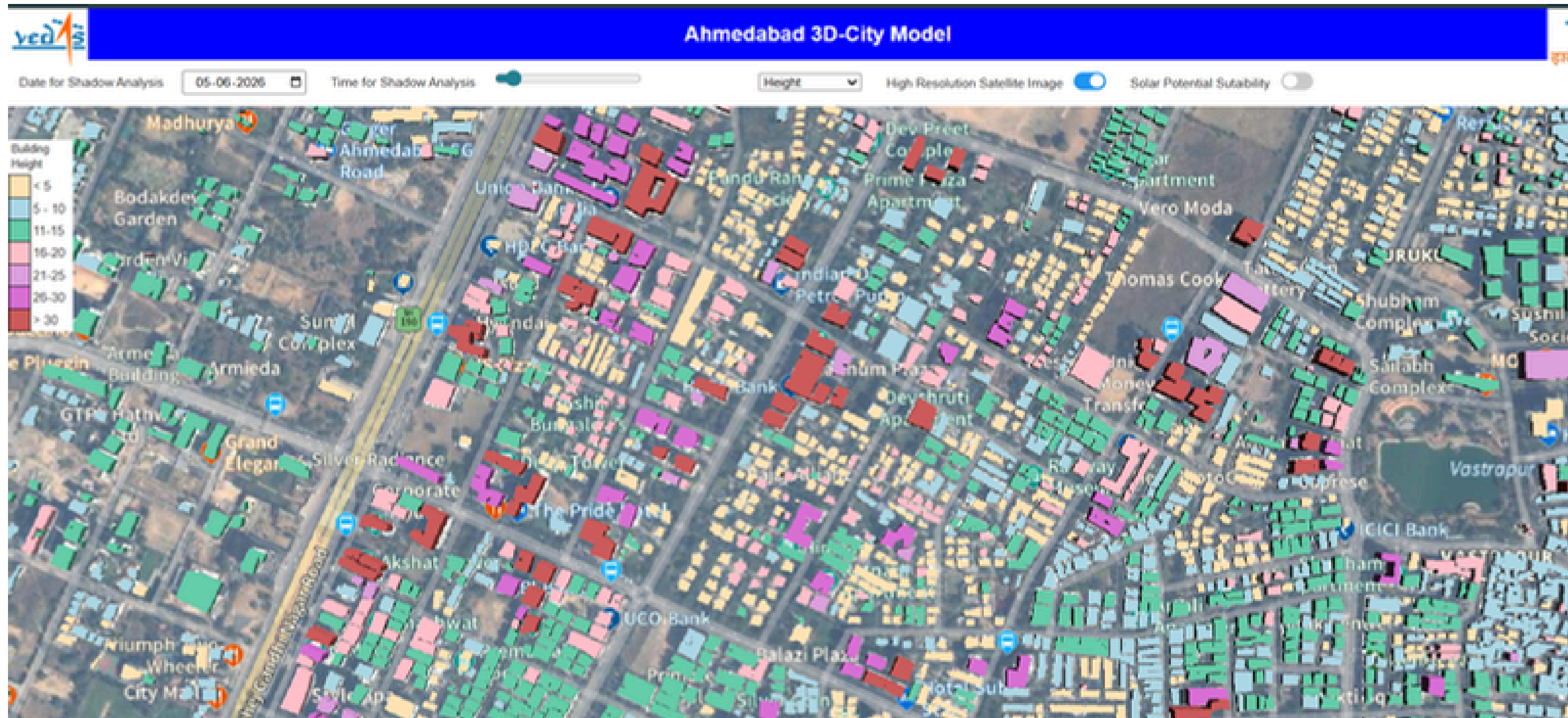


Validation

Methods Explored

- OpenStreetMap (building:levels)
- Ahmedabad 3D City Model
- Google Earth Pro 3D Visualization
- VEDAS

Explored the Ahmedabad 3D City Model platform available through VEDAS to understand building height distribution and urban morphology in Ahmedabad. The platform provides color-coded visualization of buildings based on height ranges along with shadow analysis and solar potential analysis. This was explored as an additional method for approximate building height estimation and validation for selected study areas.



Building Height Visualization Using Ahmedabad 3D City Model (VEDAS)

RESULTS

- Ahmedabad residential building datasets
- Building footprint maps
- Attribute tables with area and coordinates
- Preliminary building height estimations
- GIS visualizations for selected regions
- Understanding of CityBES dataset preparation workflows

LEARNINGS

- GIS helps visualize urban-scale building information effectively
- Building datasets can be rapidly prepared using QGIS and OpenStreetMap
- CityBES enables energy benchmarking and retrofit evaluation
- Retrofit analysis can support sustainable urban planning

CHALLENGES FACED

- Preparation of Manual Data Set and calculating area manually.
- Missing building-level information
- Difficulty in obtaining exact building heights
- Managing large spatial datasets

SOLUTIONS

- Practical QGIS exploration
- Use of multiple GIS platforms
- Cross-verification using Google Earth Pro and 3D models

FUTURE WORK

- **Advanced CityBES simulations**
- **Building energy modelling**
- **Energy analysis of buildings**
- **Ahmedabad building energy benchmarking**
- **Urban sustainability assessment**
- **Building performance evaluation**
- **Net-zero and low-carbon building design**
- **Renewable energy integration in buildings**
- **Green building and energy audit practices**

Further learning will focus on improving building energy performance, sustainability assessment, and green building implementation at both building and city scales.

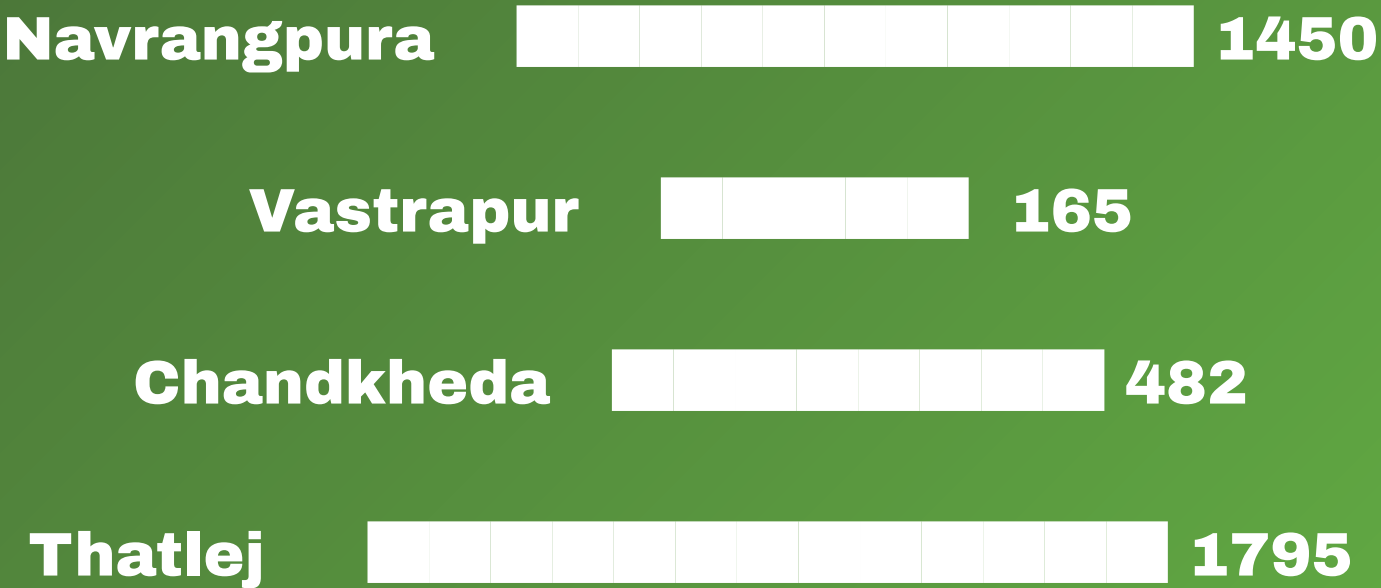
**THANK
YOU**

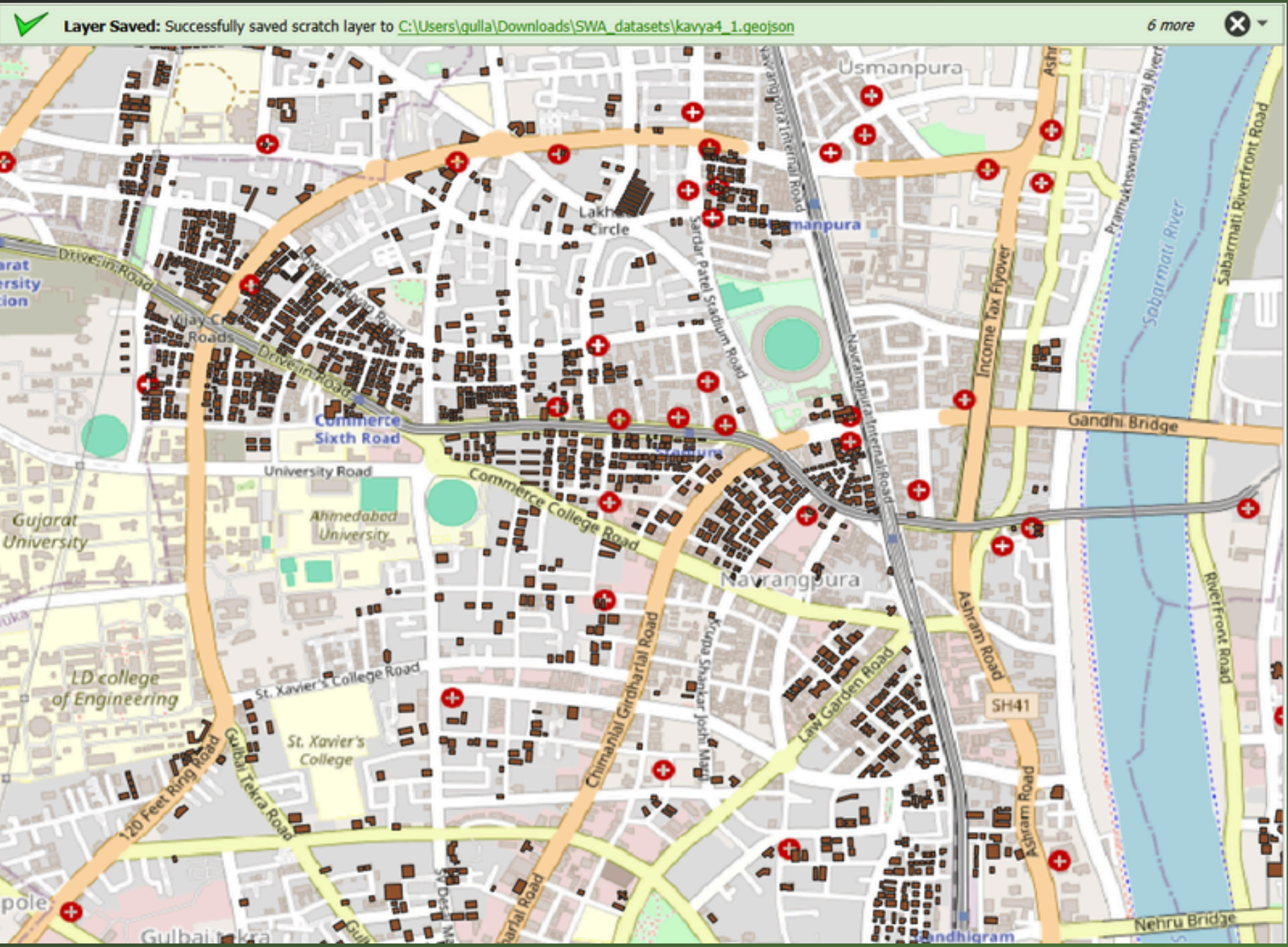


CityBES Dataset Preparation for Ahmedabad Residential Buildings

Selected Residential Study Areas --

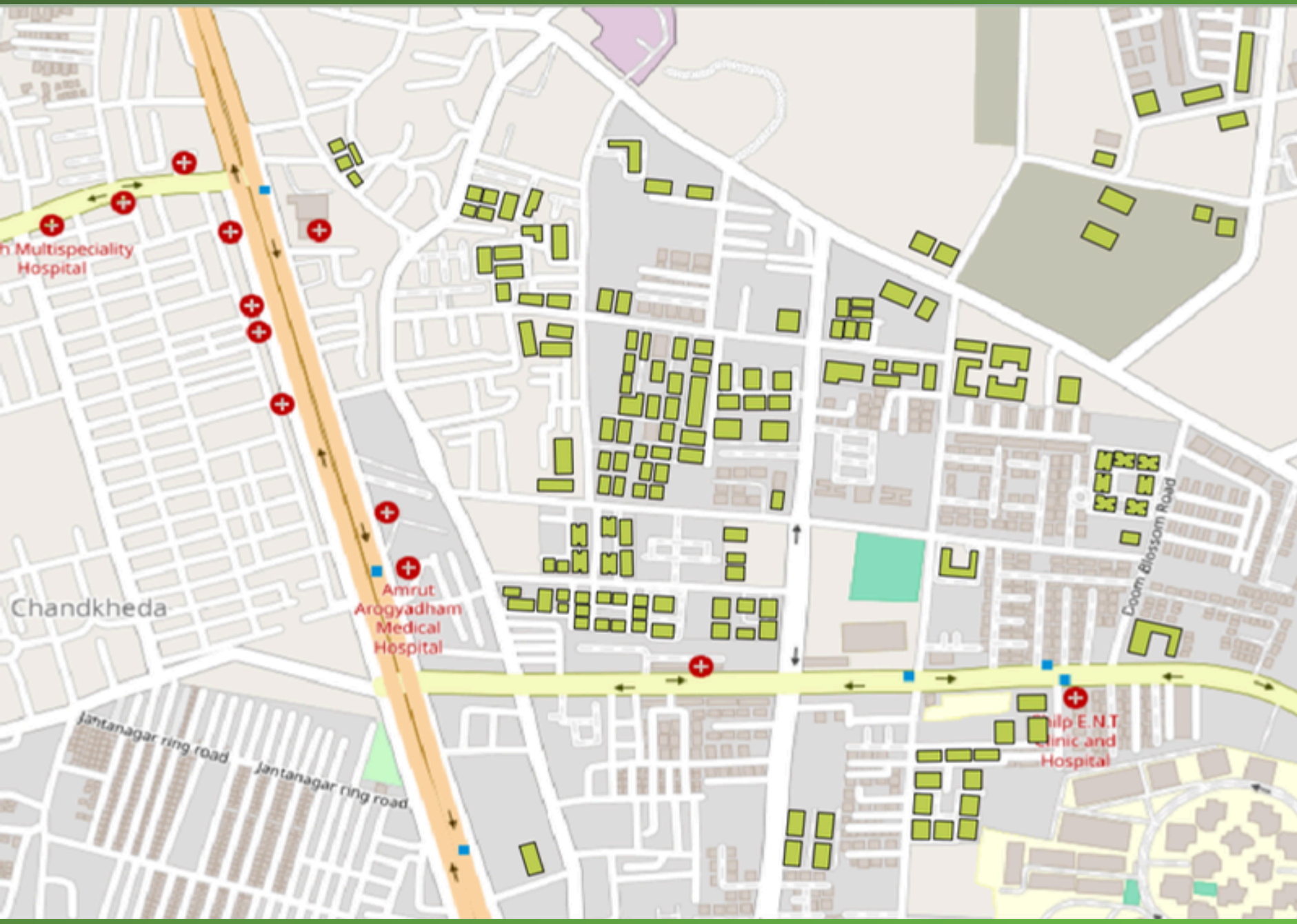
Location	Residential Buildings
Navrangpura	1450
Vastrapur	165
Chandkheda	742
Thatleji	1795

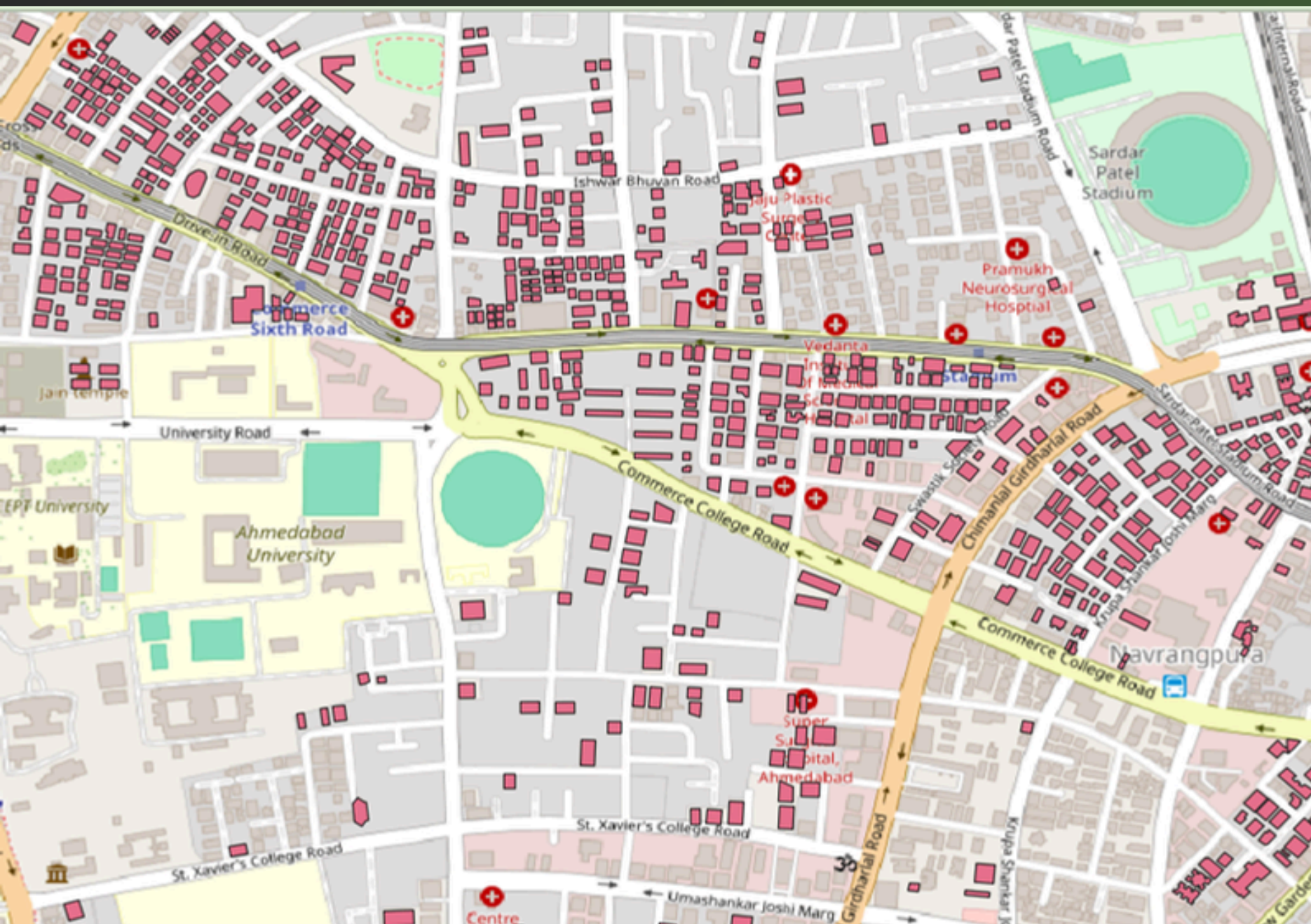




Navrangpura

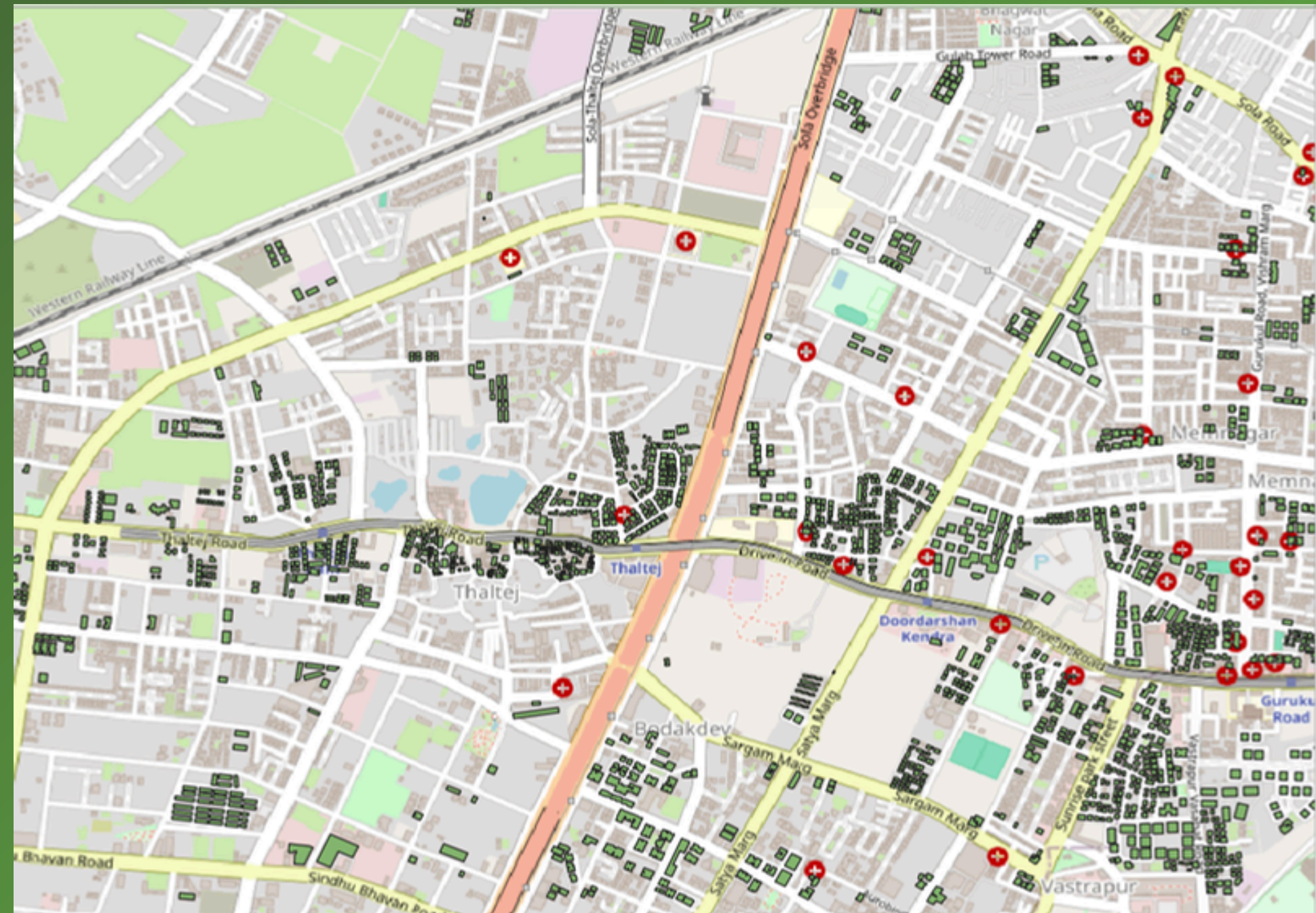
Vastrapur





Chandkheda

Thatlej



Weather Dataset for Energy Simulation

TMYx Weather File

Period of Record – 2009–2023

Selected for CityBES simulations because it represents the most recent available Typical Meteorological Year (TMYx) dataset for Ahmedabad.

[illegible]

SWA Consultancy

Dataset Ready for CityBES

✓ Residential Buildings Filtered

✓ GeoPackage Created

✓ GeoJSON Exported

✓ Latitude & Longitude
Generated

✓ Building Levels Verified

✓ Built Year Added

✓ EPW Weather File Selected

Next Steps

→ Add built year data for each area

→ Import datasets into CityBES

→ Run baseline simulation

→ Perform retrofit analysis



THANK YOU